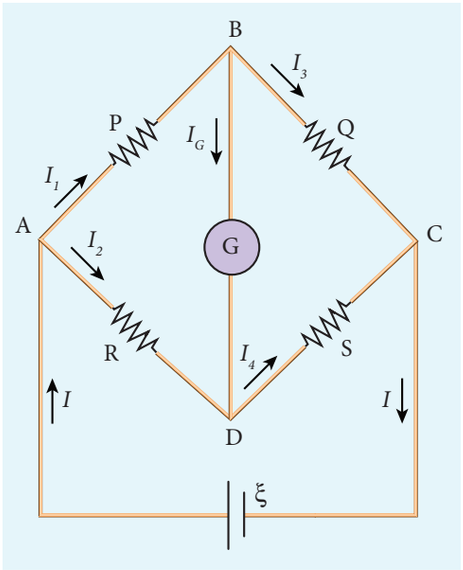


Figure 2.25 Wheatstone's bridge

Applying Kirchhoff's voltage rule to loop ABDA,

$$I_1P + I_GG - I_2R = 0 \qquad (2.47)$$

Applying Kirchhoff's voltage rule to loop ABCDA,

$$I_1P + I_3Q - I_4S - I_2R = 0 \qquad (2.48)$$

When the points B and D are at the same potential, the bridge is said to be balanced. As there is no potential difference between B and D, no current flows through galvanometer ($I_G = 0$). Substituting $I_G = 0$ in equation (2.45), (2.46) and (2.47), we get

$$I_1 = I_3 \qquad (2.49)$$

$$I_2 = I_4 \qquad (2.50)$$

$$I_1P = I_2R \qquad (2.51)$$

Substituting the equation (2.49) and (2.50) in equation (2.48)

$$I_1P + I_1Q - I_2S - I_2R = 0$$

$$I_1 (P + Q) = I_2 (R + S) \qquad (2.52)$$

Dividing equation (2.52) by equation (2.51), we get

$$\frac{P+Q}{P} = \frac{R+S}{R}$$

$$1+\frac{Q}{P} = 1+\frac{S}{R}$$

$$\frac{Q}{P} = \frac{S}{R}$$

$$\frac{P}{Q} = \frac{R}{S} \qquad (2.53)$$

This is the bridge balance condition. Only under this condition, galvanometer shows null deflection. Suppose we know the values of two adjacent resistances, the other two resistances can be compared. If three of the resistances are known, the value of unknown resistance (fourth one) can be determined.



A galvanometer is an instrument used for detecting and measuring even very small electric currents. It is extensively useful to compare the potential difference between various parts of the circuit.

EXAMPLE 2.23

In a Wheatstone's bridge P = 100 Ω, Q = 1000 Ω and R = 40 Ω. If the galvanometer shows zero deflection, determine the value of S.